

12.736

AI25BTECH11034 - Sujal Chauhan

November 3, 2025

Problem

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If $\mathbf{A} = \begin{pmatrix} 10 & 2k + 5 \\ 3k - 3 & k + 5 \end{pmatrix}$ is a symmetric matrix, then value of k is .

Property of symmetric matrices:

$$\mathbf{A} = \mathbf{A}^T \quad (1)$$

$$\mathbf{A} = \begin{pmatrix} 10 & 2k+5 \\ 3k-3 & k+5 \end{pmatrix} \quad (2)$$

$$\mathbf{A}^T = \begin{pmatrix} 10 & 3k-3 \\ 2k+5 & k+5 \end{pmatrix} \quad (3)$$

According to symmetry property:

$$\begin{pmatrix} 10 & 2k+5 \\ 3k-3 & k+5 \end{pmatrix} = \begin{pmatrix} 10 & 3k-3 \\ 2k+5 & k+5 \end{pmatrix} \quad (4)$$

Equating off-diagonal elements:

$$2k+5 = 3k-3 \quad (5)$$

$$k = 8 \quad (6)$$